

1. System Actors and Roles Overview

Role	Description	Attribution
Étudiant	The primary user who owns a workspace for their PFE. They input project details, track progress, and communicate with their supervisor.	Auto-registration via Google OAuth (institutional email only).
Encadrant (Prof)	A teacher or professional supervising one or more PFEs. Responsible for validating milestones, providing feedback, and evaluating.	Assigned by the Super Admin.
Chef de filière	Department head responsible for validating subjects, overseeing all PFEs in their department, and organizing defenses.	Assigned by the Super Admin.
Super Admin	System administrator with full control over the platform, user management, role attribution, and global oversight.	Account created during platform setup.

2. Student Workspace

After logging in, the student accesses their personal workspace with the following features:

2.1. PFE Fiche

- **PFE Title**
- **Host Organization:** Name, sector, address, and contact details.
- **Dates:** Approximate start and end dates of the internship.
- **Information:** Field of study, academic supervisor, and professional (company) supervisor.
- **Status:** Subject status (Draft, Submitted, Validated, Refused with reason).

2.2. Progress Tracking

- **Timeline:** Key milestones of the PFE (Specifications, Design, Implementation, Writing, Defense).

- **Reports:** Submission of periodic (weekly or monthly) progress reports.
- **Progress Bar:** A global progress bar visible to both the student and supervisor.
- **Logbook:** Timestamped journal or notes for follow-up.

2.3. Supervision

- **Messaging:** Private discussion thread with the supervisor.
- **Meetings:** Request an appointment or meeting slot.
- **History:** Consultation of all past feedback and comments from the supervisor.

2.4. Documents

- **Upload:** Submission of the PFE report (successive versions), internship agreement, and other deliverables.
- **Automatic Verification:** Built-in check for file format and size compliance.

2.5. Notifications

- **Deadlines:** Reminders for upcoming deadlines (report submission, defense date, pending validations).
- **Alerts:** Notifications for new messages, feedback, and status changes.

2.6. Defense

- **Consultation:** View the date, time, room, and jury composition once scheduled.
- **Upload:** Submit the final version of the report and the presentation support.

3. Supervisor (Encadrant) Workspace

This role provides tools for managing and mentoring assigned students.

- **Student List:** A list of supervised students with a synthetic progress status (color-coded/percentage).
- **Subject Validation:** Ability to validate or request modifications to the subject proposed by the student.
- **Document Review:** Consultation of reports and documents submitted, with the ability to leave comments and annotations.

- **Messaging:** Dedicated direct messaging channel for each supervised student.
 - **Evaluation Grid:** A structured evaluation form to be filled out at the end of the PFE.
 - **Automatic Alerts:** Receive alerts if a student is inactive (e.g., no submission for X days).
-

4. Department Head (Chef de filière) Workspace

This role is responsible for the overall management and oversight of all PFEs in a specific department.

- **Global Dashboard:** An overview of all PFEs in the department, filterable by status, supervisor, and progress.
 - **Validations:** Final validation of PFE subjects and internship agreements.
 - **Supervisor Assignment:** Assignment or reassignment of academic supervisors.
 - **Defense Planning:** Schedule defense dates, rooms, and jury compositions; send out summons.
 - **Reports & Statistics:** Generate statistics and reports on progress rates, late PFEs, and distribution by sector.
 - **Data Export:** Export defense reports, lists, and schedules in PDF/Excel format.
-

5. Super Admin Workspace

This role is dedicated to the overall administration and maintenance of the platform.

- **User Management:** Activate, deactivate, or delete user accounts.
- **Role Management:** Assign or revoke the "Supervisor" and "Department Head" roles.
- **Platform Configuration:** Manage academic settings (departments, academic years, PFE campaign dates, milestones).
- **Global Supervision:** An aggregated multi-department oversight dashboard.
- **System Settings:** Customize notifications, document models, and business rules (deadlines, accepted formats).

- **Audit & Monitoring:** View the audit log and technical supervision (logs, errors, backups).
-

6. Cross-Role Features (Common to All)

Certain features are integrated into the platform for all users, adapting to their specific role.

- **Role-Based Dashboard:** A synthetic view adapted to each role, showing key performance indicators (KPIs), alerts, and pending tasks.
- **Multi-Channel Notifications:** In-app and email notifications for deadlines, validations, and messages.
- **Internship Agreement Management:** Semi-automatic generation of the agreement based on the data entered.
- **Subject Library:** A library of subjects offered by companies or supervisors for students to browse and pre-select.
- **Defense Planning:** A shared calendar, room booking, and automatic generation of summons and official reports.
- **Digital Evaluation Grids:** Structured forms (for supervisor, company, jury) with automatic calculation of the final grade.
- **Document Generation:** Export of agreements, certificates, defense reports, and progress reports in PDF format.
- **Statistics & Reporting:** View success rates, average timelines, and distribution by sector/company.
- **Internal Messaging:** Project-specific discussion threads to avoid email dispersion.
- **Similarity Detection:** Anti-plagiarism check of submitted reports (integration with an external service).
- **Responsive Design:** A mobile/tablet-friendly interface for quick consultation.
- **Multi-Year Archiving:** Archiving of PFEs for future reference.
- **Multilingual Accessibility:** Interface in French and Arabic.